Material Safety Data Sheet
N-Amyl Acetate, 98%

ACC# 96281

Section 1 - Chemical Product and Company Identification

MSDS Name: N-Amyl Acetate, 98%
Catalog Numbers: AC149180000, AC149180010, AC149180050, AC149182500
Synonyms: Acetic acid, pentyl ester; Acetic acid, amyl ester; n-Amyl acetate; Amyl acetic ester; Pent-acetate; Pent-acetate 28; 1-Pentanol acetate; Pentyl acetate; n-Pentyl acetate; 1-Pentyl acetate; Primary amyl acetate
Company Identification:

Acros Organics N.V.
One Reagent Lane
Fair Lawn, NJ 07410

For information in North America, call: 800-ACROS-01
For emergencies in the US, call CHEMTREC: 800-424-9300

Section 2 - Composition, Information on Ingredients

CAS#	Chemical Name	Percent	EINECS/ELINCS
628-63-7	n-Amyl acetate	ca.100	211-047-3

Hazard Symbols: F
Risk Phrases: 10

Section 3 - Hazards Identification

EMERGENCY OVERVIEW

Appearance: colorless liquid. Flash Point: 23 deg C. **Warning! Flammable liquid and vapor.** May cause central nervous system depression. May cause liver damage. May cause cardiac disturbances. Causes eye irritation. May cause skin irritation. May cause digestive tract irritation.
Target Organs: Heart, central nervous system, liver.

Potential Health Effects

Eye: Causes eye irritation. May cause chemical conjunctivitis and corneal damage.
Skin: May cause skin irritation. May be absorbed through the skin. May cause cyanosis of the extremities.
Ingestion: May cause gastrointestinal irritation with nausea, vomiting and diarrhea. May cause effects similar to those for inhalation exposure. Ingestion of large amounts may cause CNS depression.
Inhalation: Inhalation of high concentrations may cause central nervous system effects characterized by nausea, headache, dizziness, unconsciousness and coma. May cause liver abnormalities. Aspiration may lead to pulmonary edema. Vapors may cause dizziness or suffocation. May cause cardiac abnormalities. Causes irritation of the mucous membrane and upper respiratory tract. May cause burning sensation in the chest.
Chronic: Prolonged or repeated skin contact may cause dermatitis. Chronic exposure may cause liver damage. Chronic exposure will cause neurological degradation and/or abnormalities.

Section 4 - First Aid Measures

Eyes: Immediately flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Get medical aid.
Skin: Get medical aid. Flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Wash clothing before reuse.
Ingestion: Do NOT induce vomiting. If victim is conscious and alert, give 2-4 cupfuls of milk or water. Never give anything t mouth to an unconscious person. Get medical aid.

Inhalation: Get medical aid immediately. Remove from exposure and move to fresh air immediately. If not breathing, give artificial respiration. If breathing is difficult, give oxygen.
Notes to Physician: Treat symptomatically and supportively.

Section 5 - Fire Fighting Measures

General Information: As in any fire, wear a self-contained breathing apparatus in pressure-demand, MSHA/NIOSH (approved or equivalent), and full protective gear. Water runoff can cause environmental damage. Dike and collect water use to fight fire. Vapors may form an explosive mixture with air. Vapors can travel to a source of ignition and flash back. Combustion generates toxic fumes. Will burn if involved in a fire. Use water spray to keep fire-exposed containers cool. Containers may explode in the heat of a fire. Flammable liquid and vapor.
Extinguishing Media: For small fires, use dry chemical, carbon dioxide, water spray or alcohol-resistant foam. For large fires, use water spray, fog, or alcohol-resistant foam. Use water spray to cool fire-exposed containers. Water may be ineffective. Do NOT use straight streams of water.
Flash Point: 23 deg C (73.40 deg F)
Autoignition Temperature: 360 deg C (680.00 deg F)
Explosion Limits, Lower:1.1 vol %
Upper: 7.5 vol %
NFPA Rating: (estimated) Health: 1; Flammability: 3; Instability: 0

Section 6 - Accidental Release Measures

General Information: Use proper personal protective equipment as indicated in Section 8.
Spills/Leaks: Absorb spill with inert material (e.g. vermiculite, sand or earth), then place in suitable container. Remove all sources of ignition. Use a spark-proof tool. Provide ventilation. A vapor suppressing foam may be used to reduce vapors.

Section 7 - Handling and Storage

Handling: Wash thoroughly after handling. Remove contaminated clothing and wash before reuse. Use only in a well-ventilated area. Ground and bond containers when transferring material. Use spark-proof tools and explosion proof equipment. Avoid contact with skin and eyes. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Avoid contact with heat, sparks and flame. Avoid ingestion and inhalation. Do not pressurize, cut, weld, braze, solder, drill, grind, or expose empty containers to heat, sparks or open flames.
Storage: Keep away from heat, sparks, and flame. Keep away from sources of ignition. Store in a cool, dry, well-ventilated area away from incompatible substances. Flammables-area.

Section 8 - Exposure Controls, Personal Protection

Engineering Controls: Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower. Use adequate general or local exhaust ventilation to keep airborne concentrations below the permissible exposure limits.
Exposure Limits

Chemical Name	ACGIH	NIOSH	OSHA - Final PELs
n-Amyl acetate	50 ppm TWA (listed under Pentyl acetate, all isomers); 100 ppm STEL (listed under Pentyl acetate, all isomers)	100 ppm TWA; 525 mg/m3 TWA 1000 ppm IDLH	100 ppm TWA; 525 mg/m3 TWA

OSHA Vacated PELs: n-Amyl acetate: 100 ppm TWA; 525 mg/m3 TWA
Personal Protective Equipment
Eyes: Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.
Skin: Wear appropriate protective gloves to prevent skin exposure.

Clothing: Wear appropriate protective clothing to prevent skin exposure.
Respirators: A respiratory protection program that meets OSHA's 29 CFR 1910.134 and ANSI Z88.2 requirements or European Standard EN 149 must be followed whenever workplace conditions warrant a respirator's use.

Section 9 - Physical and Chemical Properties

Physical State: Liquid
Appearance: colorless
Odor: odor of pears - banana-like
pH: Not available.
Vapor Pressure: 4 mm Hg @ 20 deg C
Vapor Density: 4.5 (air=1)
Evaporation Rate:0.42 (butyl acetate=1)
Viscosity: 0.91 cps @ 22 deg C
Boiling Point: 142 deg C
Freezing/Melting Point:-70.8 deg C
Decomposition Temperature: Not available.
Solubility: Slightly soluble.
Specific Gravity/Density:0.8760 (water=1)
Molecular Formula:C7H14O2
Molecular Weight:130.0968

Section 10 - Stability and Reactivity

Chemical Stability: Stable under normal temperatures and pressures.
Conditions to Avoid: High temperatures, incompatible materials, ignition sources, excess heat.
Incompatibilities with Other Materials: Strong oxidizing agents, strong bases.
Hazardous Decomposition Products: Carbon monoxide, irritating and toxic fumes and gases, carbon dioxide.
Hazardous Polymerization: Has not been reported.

Section 11 - Toxicological Information

RTECS#:
CAS# 628-63-7: AJ1925000
LD50/LC50:
CAS# 628-63-7:
Oral, rabbit: LD50 = 7400 mg/kg;
Oral, rat: LD50 = >1600 mg/kg;
Carcinogenicity:
CAS# 628-63-7: Not listed by ACGIH, IARC, NIOSH, NTP, or OSHA.
Epidemiology: No information available.
Teratogenicity: No information available.
Reproductive Effects: No information available.
Neurotoxicity: No information available.
Mutagenicity: No information available.
Other Studies: Standard Draize Test: Administration into the eye = 300 ppm.

Section 12 - Ecological Information

Ecotoxicity: Fish: Bluegill/Sunfish: LC50 = 650 mg/L; 96 Hr; Static bioassay at 23°C Mosquito Fish: LC50 = 65 mg/L; 24-9(Hr; Unspecified If released on land or in water, volatilization would be important (half-life 5.9 hr in a typical river) and biodegradation, should be a dominant degradative process. Adsorption to soil or sediment would not occur to any significant extent, so leaching into groundwater may occur. Some chemical hydrolysis may occur but only under fairly alkaline conditions. n-Amyl acetate would not be expected to bioconcentrate in aquatic organism.

Environmental: In air, n-amyl acetate will be scavenged by rain and degrade by reaction with photochemically produced hydroxyl radicals estimated half-life 4.5 days).
Physical: No information available.
Other: No information available.

Section 13 - Disposal Considerations

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. US EPA guidelines for the classification determination are listed in 40 CFR Parts 261.3. Additionally, waste generators must consult state and local hazardous waste regulations to ensure complete and accurate classification.
RCRA P-Series: None listed.
RCRA U-Series: None listed.

Section 14 - Transport Information

	US DOT	IATA	RID/ADR	IMO	Canada TDG
Shipping Name:	AMYL ACETATES				No information available.
Hazard Class:	3				
UN Number:	UN1104				
Packing Group:	III				

Section 15 - Regulatory Information

US FEDERAL

TSCA
CAS# 628-63-7 is listed on the TSCA inventory.
Health & Safety Reporting List
None of the chemicals are on the Health & Safety Reporting List.
Chemical Test Rules
None of the chemicals in this product are under a Chemical Test Rule.
Section 12b
CAS# 628-63-7: 4/12b
TSCA Significant New Use Rule
No of the chemicals in this material have a SNUR under TSCA.
SARA

CERCLA Hazardous Substances and corresponding RQs
CAS# 628-63-7: 5000 lb final RQ; 2270 kg final RQ
SARA Section 302 Extremely Hazardous Substances
None of the chemicals in this product have a TPQ.
SARA Codes
CAS # 628-63-7: acute, flammable.
Section 313
No chemicals are reportable under Section 313.
Clean Air Act:
This material does not contain any hazardous air pollutants. This material does not contain any Class 1 Ozone depletors. This material does not contain any Class 2 Ozone depletors.
Clean Water Act:
CAS# 628-63-7 is listed as a Hazardous Substance under the CWA. None of the chemicals in this product are listed as Priorit Pollutants under the CWA. None of the chemicals in this product are listed as Toxic Pollutants under the CWA.
OSHA:
None of the chemicals in this product are considered highly hazardous by OSHA.
STATE
CAS# 628-63-7 can be found on the following state right to know lists: California, New Jersey, Pennsylvania, Minnesota,

Massachusetts.
California No Significant Risk Level: None of the chemicals in this product are listed.

European/International Regulations
European Labeling in Accordance with EC Directives
Hazard Symbols:
F
Risk Phrases:
R 10 Flammable.

Safety Phrases:
S 16 Keep away from sources of ignition - No smoking.
S 33 Take precautionary measures against static discharges.
S 9 Keep container in a well-ventilated place.
S 23C Do not breathe vapour.

WGK (Water Danger/Protection)
CAS# 628-63-7: 1
Canada - DSL/NDSL
CAS# 628-63-7 is listed on Canada's DSL List.
Canada - WHMIS
This product does not have a WHMIS classification.
Canadian Ingredient Disclosure List
CAS# 628-63-7 is listed on the Canadian Ingredient Disclosure List.
Exposure Limits

Section 16 - Additional Information

MSDS Creation Date: 6/08/1999
Revision #3 Date: 3/18/2003

The information above is believed to be accurate and represents the best information currently available to us. However, we make no warranty of merchantabilit or any other warranty, express or implied, with respect to such information, and we assume no liability resulting from its use. Users should make their own investigations to determine the suitability of the information for their particular purposes. In no event shall Fisher be liable for any claims, losses, or damages of any third party or for lost profits or any special, indirect, incidental, consequential or exemplary damages, howsoever arising, even if Fisher has been advised of the possibility of such damages.